DEPARTMENT OF
COMPUTER SCIENCE & ENGINEERING

Experiment 5

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Branch: CSE
Section/Group: KRG 2-A
Semester: 6th
Date of Performance:26/02/2026
Subject Name: System Design
Subject Code: 23CSH-314

1. Aim: Design a Messenger Application(Similar to WhatsApp / Facebook Messenger): To design and analyze the architecture of a scalable real-time messenger application that supports one-to-one and group communication with high availability and low latency.

2. Objective:

- To implement real-time message delivery using WebSockets.
- To support one-to-one and group messaging.
- To preserve chat history.
- To ensure high availability and reliability.
- To design APIs for user management and chat operations.
- To design the system for large-scale usage (up to 1 Billion users).

3. Tools Used:

- Frontend (ReactJS, HTML, CSS, JavaScript, WebSocket) o Built responsive UI with real-time communication support
- Backend (Node.js / Java / Spring Boot) o Developed scalable REST APIs & WebSocket server
- Database (MongoDB / MySQL / PostgreSQL) o Managed structured & unstructured data efficiently
- Redis o Implemented caching & pub-sub for high performance
- Deployment (AWS / GCP, Load Balancer, Docker) o Cloud deployment with scalability, containerization & traffic distribution

4. System Requirements:

A. Functional Requirements

- User Registration
  a. POST /user/register
  b. Fields: userName, email, phoneNumber
- User Login

   a. POST /user/login
- One-to-One Messaging
   a. WS: /messages/send

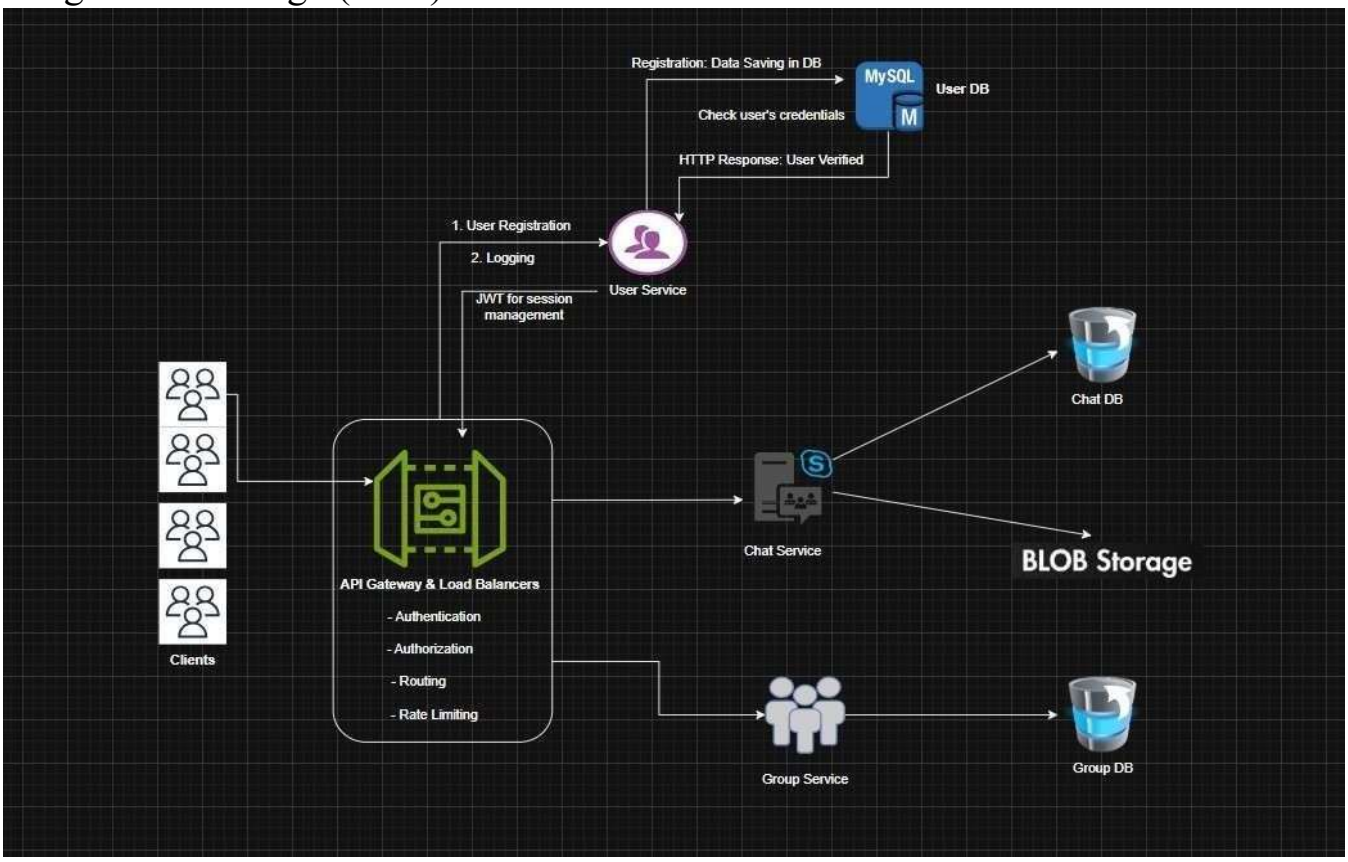
   b. GET /messages/{userId}/{receiverId}
   c. Pagination support
- Group Messaging
   a. POST /groups/create
   b. POST /groups/{groupId}/add
   c. DELETE /groups/{groupId}/remove
   d. WS: /messages/send
- Message Types
   a. Text messages
   b. Media messages (images, videos, files)
- Message History
   a. Persistent storage
   b. Chat list view
- Read Receipts
   a. Delivered
   b. Seen

B. Non-Functional Requirements

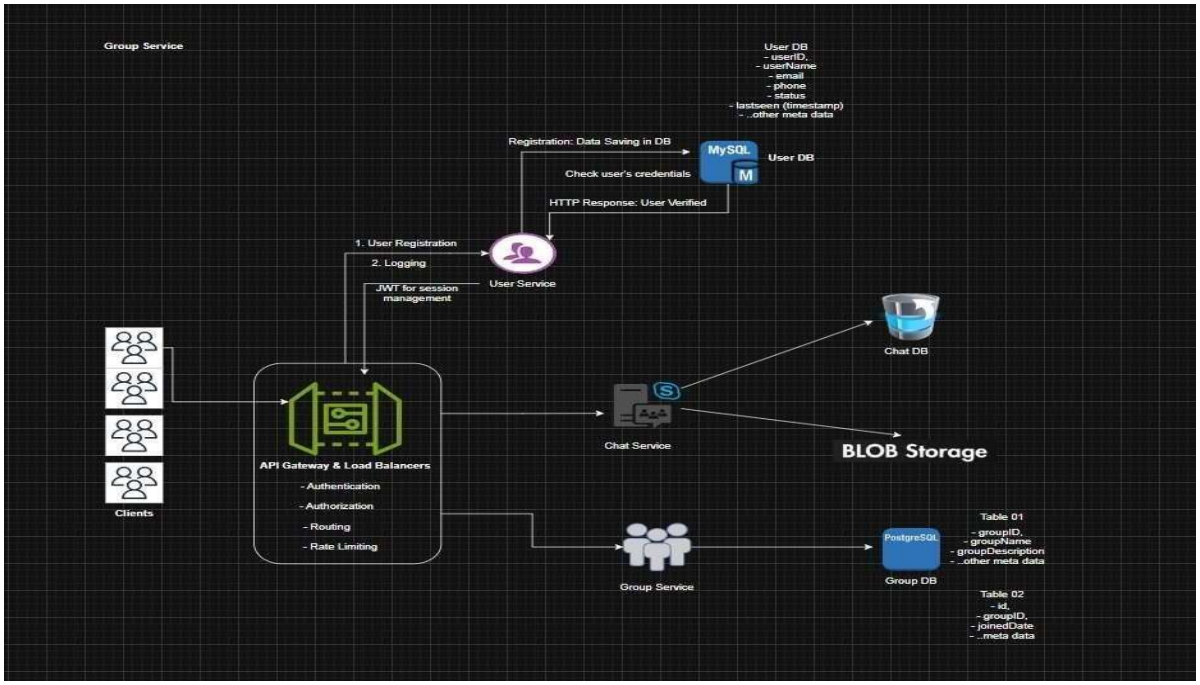
- Scalability o Target: 1 Billion users o 100 messages per user per day o ≈ 100 Billion messages/day o Estimated storage: ~100 TB

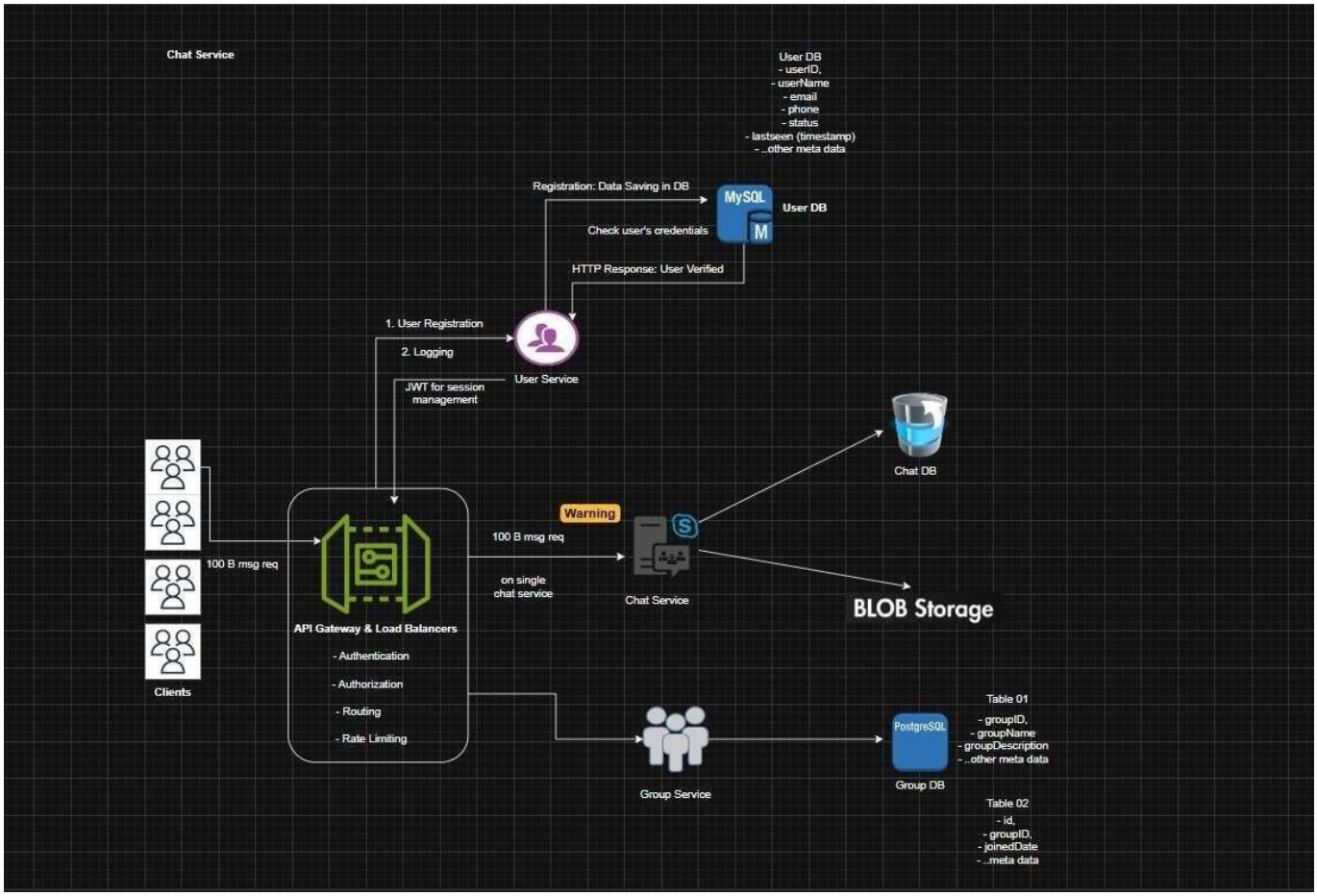
- CAP Theorem
  System prioritizes: o
  High Availability o
  Partition Tolerance o
  Eventual Consistency
- Latency: 200-300ms o Target message delivery time: 200–300 ms

5. High Level Design (HLD)



6. Low Level Design (LLD):


Group Service
User DB
- userID,
- userName
- email
- phone
- status
- lastseen (timestamp)
- ..other meta data
Registration: Data Saving in DB
MySQL
User DB
Check user's credentials
HTTP Response: User Verified
1. User Registration
2. Logging
User Service
JWT for session management
Chat DB
Clients
API Gateway & Load Balancers
- Authentication
- Authorization
- Routing
- Rate Limiting
Chat Service
BLOB Storage
Table 01
- groupID,
- groupName
- groupDescription
- ..other meta data
PostgreSQL
Group Service
Group DB
Table 02
- id,
- groupID,
- joinedDate
- ..meta data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JWT for session management
Chat DB
Warning
100 B msg req
on single chat service
Chat Service
BLOB Storage
100 B msg req
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- Authorization
- Routing
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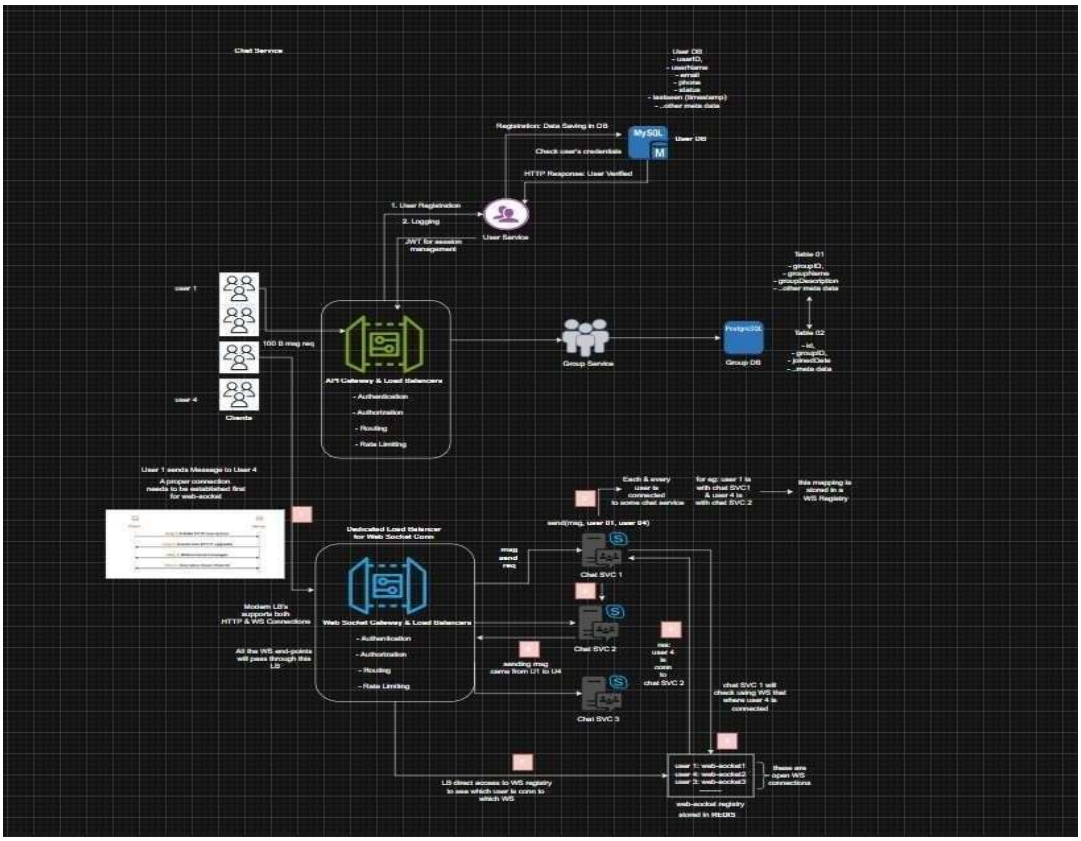

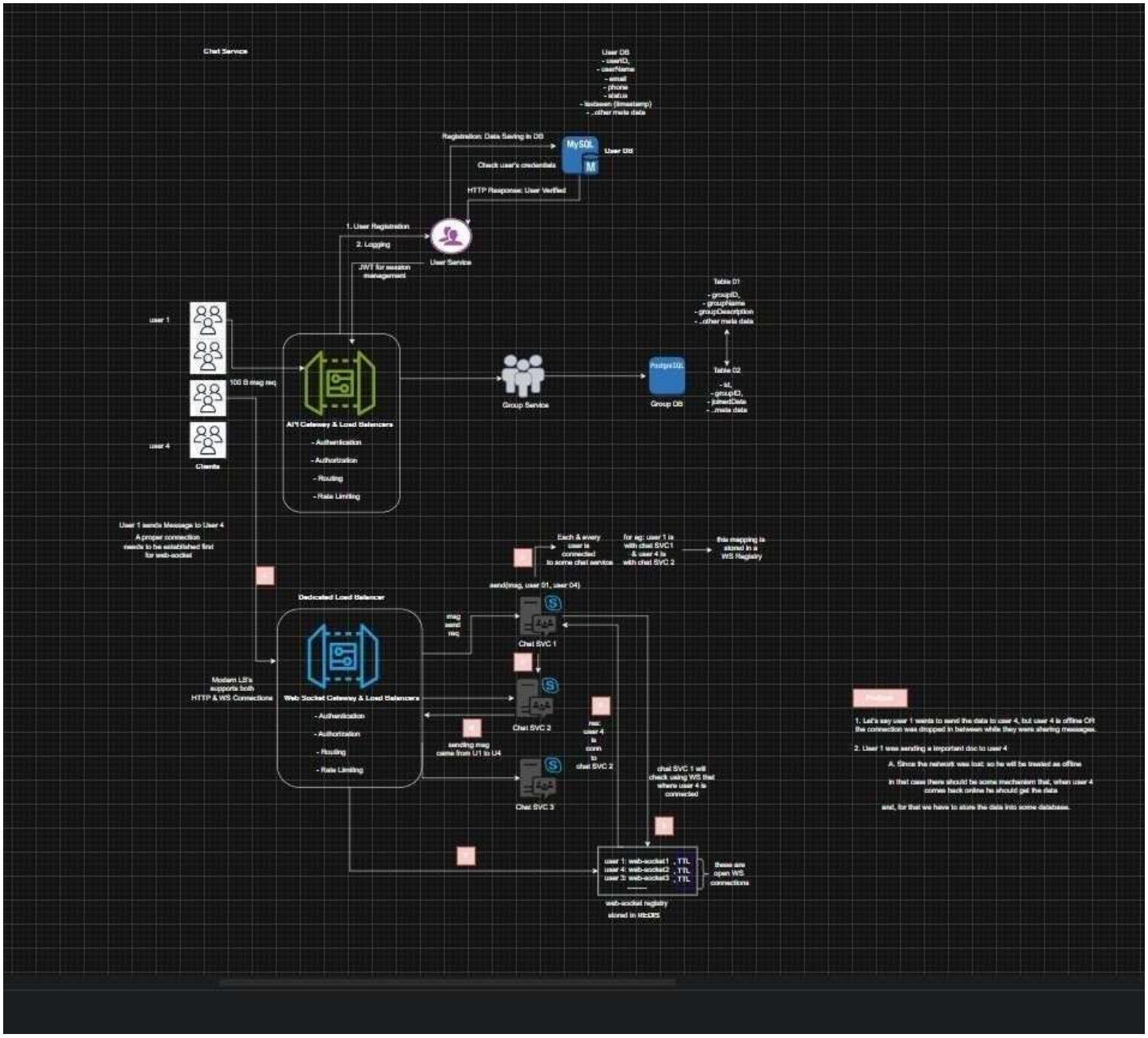
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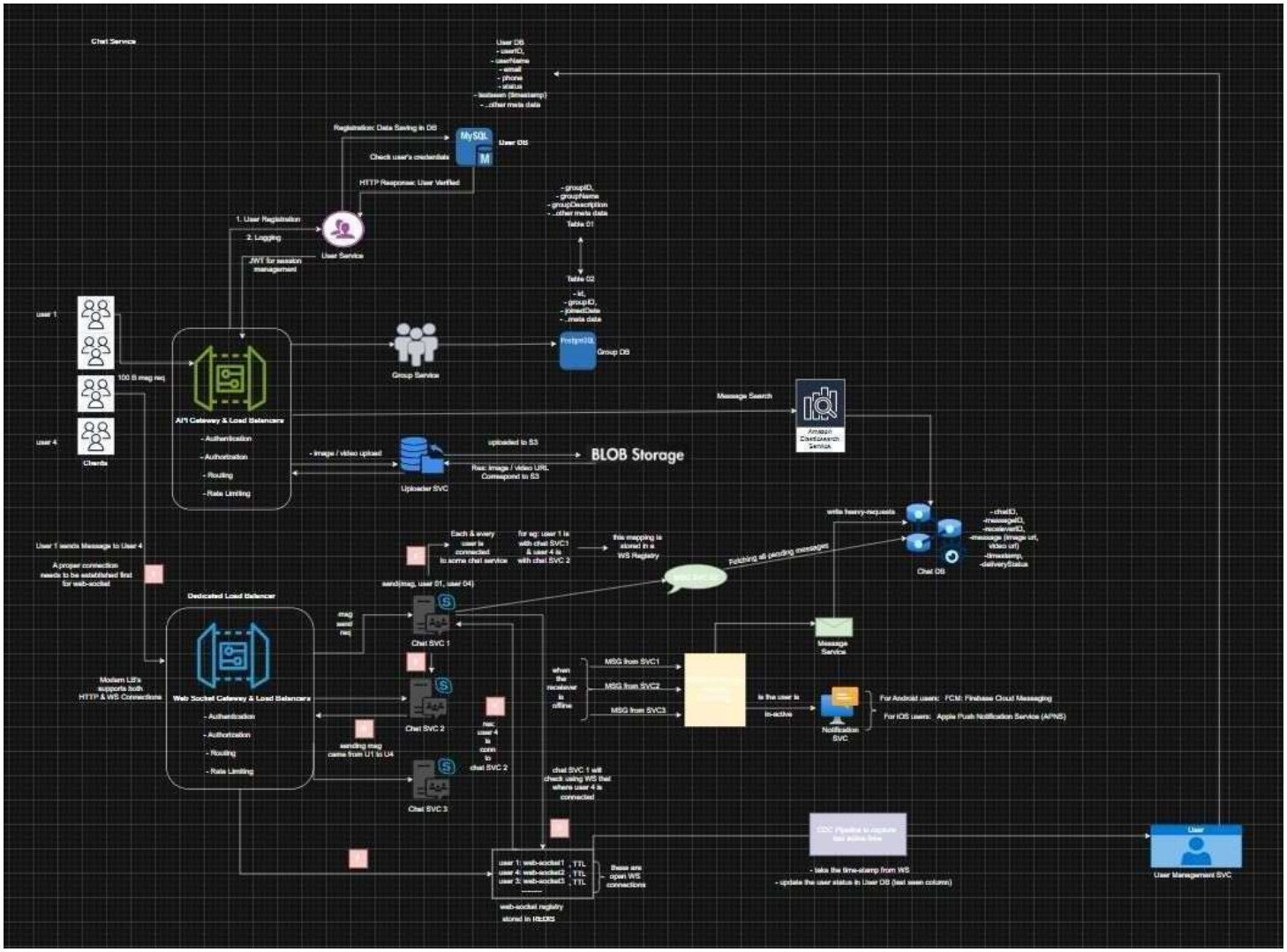
- ..other meta data
Table 02
- id,
- groupID,
- joinedDate
- ..meta data
user 1
100 B msg req
user 4
Clients
API Gateway & Load Balancers
- Authentication
- Authorization
- Routing
- Rate Limiting
Group Service
Group DB
User 1 sends Message to User 4
A proper connection needs to be established first for web-socket
Each & every user is connected to some chat service
for eg: user 1 is with chat SVC1 & user 4 is with chat SVC 2
this mapping is stored in a WS Registry
send(msg, user 01, user 04)
Dedicated Load Balancer
msg sent req
Chat SVC 1
Modern LB's supports both HTTP & WS Connections
Web Socket Gateway & Load Balancers
- Authentication
- Authorization
- Routing
- Rate Limiting
Chat SVC 2
req: user 4 is conn to chat SVC 2
sending msg came from U1 to U4
Chat SVC 3
chat SVC 1 will check using WS that where user 4 is connected
user 1: web-socket1 , TTL
user 4: web-socket2 , TTL
user 3: web-socket3 , TTL
these are open WS connections
web-socket registry stored in REDIS
1. Let's say user 1 wants to send the data to user 4, but user 4 is offline OR the connection was dropped in between while they were sharing messages.
2. User 1 was sending a important doc to user 4
A. Since the network was lost so he will be treated as offline
In that case there should be some mechanism that, when user 4 comes back online he should get the data
and, for that we have to store the data into some database.



7. Scalability Solution

- Horizontal Scaling
  Add multiple application servers behind a load balancer to distribute traffic and handle billions of requests efficiently.
- Database Sharding
  Partition data based on userID or chatID so that message load is distributed across multiple database servers.
- Caching with Redis
  Store frequently accessed chats and recent messages in Redis to reduce database load and improve response time.
- Message Queue System
  Use Kafka/RabbitMQ for asynchronous message processing and reliable message delivery.
- CDN for Media Files
  Use Content Delivery Network to efficiently deliver images, videos, and files globally with low latency.

8. Learning Outcomes (What I Have Learnt)

- Understand real-time communication using WebSockets.
- Learn how to design scalable distributed systems.
- Apply CAP Theorem concepts in real-world system design.
- Design RESTful APIs and WebSocket endpoints.
- Analyze trade-offs between consistency, availability, and latency.